# Questions by topic: Mineral resources (the lithosphere) - Set 1 (75 marks)

**Answer all questions in the spaces provided.**

**0 1 . 1** Describe how hydrothermal deposition produces exploitable concentrations of metal ores.

**[4 marks]**

**0 1 . 2** Explain how Proterozoic marine sediments produced the world's major iron ore deposits.

**[2 marks]**

**0 1 . 3** Explain how alluvial (physical sediment) processes separate and concentrate minerals such as gold and tin ore.

**[2 marks]**

**0 1 . 4** Give **two** examples of mineral deposits formed by biological sedimentation and state their origin.

**[2 marks]**

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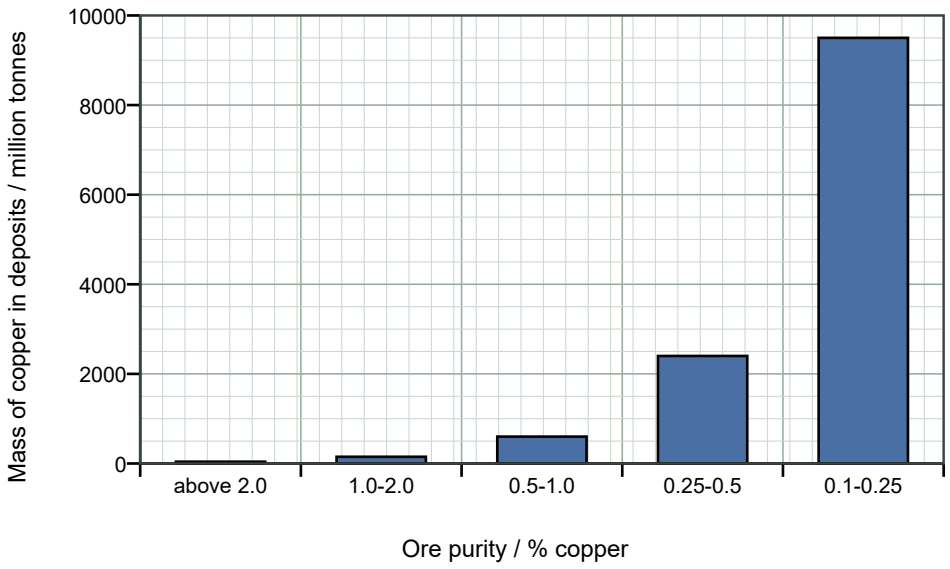
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0 2 Figure 1 shows the mass of copper contained in the Earth's crust at different ore purities, illustrating Lasky's principle.

**Figure 1**




0 2 . 1 State Lasky's principle.

**[1 mark]**

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0 2 . 2 Use Figure 1 to calculate how many times more copper is contained in deposits of 0.25-0.5% purity than in deposits above 2.0%.

**[1 mark]**

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**Answer** [ ] times

**0 2 . 3** Explain the difference between the **reserves** and the **resource** of a mineral.

**[3 marks]**

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**0 2 . 4** Explain, using Lasky's principle, why the ability to exploit slightly lower grade ores greatly increases reserves.

**[2 marks]**

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**0 2 . 5** Explain why the main problem for future mineral supplies is not the quantity that exists in the crust.

**[3 marks]**

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**0 3** Table 1 shows some techniques used to locate mineral deposits.

**0 3 . 1** Complete Table 1 by adding the name of the technique or a description of how it works.

**[4 marks]**

**Table 1**

| Technique | How it works |
|---|---|
| Gravimetry | |
| | Detects rocks that are more magnetic, such as magnetite iron ore or ores of tungsten and cobalt |
| Seismic survey | |
| | Measures how difficult it is for electricity to pass through rocks; sedimentary rocks with high water content have lower values than igneous rocks |

**0 3 . 2** Explain why trial drilling is usually carried out only after remote sensing surveys.

**[3 marks]**

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**0 3 . 3** Explain how infrared spectroscopy from satellites can help identify mineral deposits.

**[2 marks]**

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**0 3 . 4** Give **one** advantage of portable field analysis equipment such as X-ray fluorescence.

**[1 mark]**

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**0 4** Table 1 shows information about two copper deposits, X and Y, being assessed for mining. The current cut-off ore grade for copper is 0.4%.

**Table 1**

| Feature | Deposit X | Deposit Y |
|---|---|---|
| Ore grade / % copper | 0.9 | 0.35 |
| Depth of deposit / m | 350 | 40 |
| Overburden | Hard igneous rock | Loose glacial sediment |
| Distance to nearest railway / km | 120 | 8 |
| Annual rainfall / mm | 2 100 | 500 |
| Chemical form | Copper sulfide | Copper sulfide |

**0 4 . 1** Explain what is meant by the **cut-off ore grade** and state which deposit is currently included in the copper reserves.

**[2 marks]**

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**0 4 . 2** Use Table 1 to discuss the factors that affect the viability of mining each deposit.

**[6 marks]**

**0 4 . 3** State **two** changes that would decrease the cut-off ore grade.

**[2 marks]**

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**0 5 . 1** Explain why mineral extraction causes more land-use conflict than most other human activities.

**[2 marks]**

**0 5 . 2** Describe how each of the following environmental impacts of an operating mine can be reduced.

**[5 marks]**

**Dust**

**Noise**

**Turbid drainage water**

**Acidic leachate from spoil**

**Loss of amenity**

**0 5 . 3** Explain how spoil heaps can be made stable and describe what happened at Aberfan in 1966.

**[3 marks]**

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**0 6** A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

**0 6 . 1** Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

**[2 marks]**

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**0 6 . 2** Explain the conditions that would need to be met for the site to be used for housing or farmland.

**[2 marks]**

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**0 6 . 3** Explain why moving the wildlife from a site before mining begins is rarely completely successful.

**[1 mark]**

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**0 7 . 1** Describe how bioleaching is used to extract copper from low-grade ores or spoil heaps, and explain why it may be more sustainable than smelting.

**[5 marks]**

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**0 7 . 2** Describe the process of phytomining.

**[3 marks]**

**0 7 . 3** Explain how polymer adsorption may provide a low-energy source of uranium in the future.

**[2 marks]**

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**0 8 . 1** Describe polymetallic (manganese) nodules and explain why they are not yet exploited on a large scale.

**[3 marks]**

**0 8 . 2** Explain the environmental impacts of mining polymetallic nodules.

**[3 marks]**

**0 8 . 3** Explain why rare earth metals are a supply concern and describe one new extraction technique.

**[4 marks]**

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**END OF QUESTIONS**

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